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Quantum Social Prescribing: Multi-Scale Quantum Algorithms, Psychoneuroimmunological Grounding, and Multi-Method Health AI Governance for Integrated Neighbourhood Care

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Portal & Cryptographic Receipts: arunnadarasa.github.io/quantum-social-prescribing | **HF Hub & Space:** hf.co/socialprescribing/quantinuum-social-prescribing

Abstract. Social prescribing connects primary care, community pharmacies, and Link Workers to Voluntary, Community, Faith, and Social Enterprise (VCFSE) pathways addressing loneliness, housing precarity, physical inactivity, and high-anticholinergic-burden polypharmacy [1, 2, 3]. While routine administrative referral checklists often rely on greedy single-label signposting, classical mathematical formalisms—including Bayesian networks, gradient-boosted decision trees (GBDTs), mixed-integer linear programming (MILP) [15], and non-Markovian survival models—already address uncertainty, cubic synergies, caseload caps, and attrition classically. Here we introduce **Quantum Social Prescribing (QSPM)**, a receipt-verified research architecture organized into three functional tiers across synthetic, theoretically parameterized cohorts and Hamiltonians: **(Tier 1)** socio-clinical optimization and multi-task triage, **(Tier 2)** phenomenological quantum-cognition [13], Zeno/Anti-Zeno [14], and open-system biomarker hypothesis generation, and **(Tier 3)** post-quantum EHR provenance and multi-method clinical safety governance. Across live **Guppy 1.0** → **HUGR** emulator jobs on **Quantinuum Nexus** `Helios-1E-lite` (6,144 shots), `aqora:nexus:H2-Emulator` (1,536 shots), and Selene (3,072 shots), coherent Bell readout achieves metrological visibility $P(00) = 0.9927$ vs. 0.3525 under mid-circuit collapse, while an 8-rung affine calibration ladder confirms emulator-to-statevector readout linearity ($R^2 = 0.99995$ on `Helios-1E-lite`; $R^2 = 0.99923$ on Selene) for a parameterized 2-qubit Zeno channel whose baseline optimal cadence $N^*_{\text{check}} = 6$ varies across a 125-point sensitivity grid (median = 4, IQR $[0, 6]$). On an 8-pathway ($n_{\text{path}} = 8$) pure 3-body pharmacy–VCFSE HUBO, **Haiqu SDK v1.7.1** compresses LR-QAOA depth from 241 → 14 (−94.19%), sampling the exact ground bundle `11110100` ($E_0 = -20.3000$) on 74.71% (765/1,024) of raw pre-postprocessing shots; on a 120-variable ($N_{\text{pat}} = 24$) Core20PLUS5 caseload QUBO, **OpenJij** single-flip annealing reaches 82.79% of the 10.25 ms HiGHS MILP optimum (29.4491), whereas 4-bit Kempe swaps recover 100.00% in 8.72 ms. On **Modal Cloud** and **Hugging Face** ($N_{\text{cohort}} = 2,400$), a 4-qubit Frobenius VQC (`QSPMLM-VQC`, 60 quantum + 99 classical readout parameters) and full-split IQP kernel ($N_{\text{test}} = 480$) both achieve 99.79% (479/480) on linearly separable archetypes, whereas on a non-linear cubic ($k = 3$) synergy cohort linear `LogisticRegression` drops to 32.71% (0.0% falls recall) while `QSPMLM-VQC` reaches 65.21% (76.92% falls recall) behind classical `GBDT` (77.92%) and `RBF-SVM` (79.58%). Tier 2 pairs an **HQS** `struqture-py/qoqo` 4-spin Lindbladian ($n_{\text{spin}} = 4$; median 76.6% simulated IL-6 suppression across a 49-point rate grid), **QProtSim** on crystal deamino-oxytocin (`PDB 1XY1`) [17, 27], **AlphaGenome** GENCODE v46 metadata, `rhyme-beta` compiler checks, and **Text2WetLab** OT-2 verification. Tier 3 integrates **NIST FIPS 204** `ML-DSA-65` FHIR R4 signatures, Kashefi Primitive Atlas simulability contracts [16], and live **TypeSafe Jev 1.14** auditing [18], which appropriately penalizes uncalibrated behavioural/Zeno clinical accuracy (1.40/5 and 0.77/5) with 0.00 autonomous deployment probability. Decoupling discrete HUBOs (solved via Rosenberg linearization + HiGHS MILP in 1.32 s at $n_{\text{path}} = 48$) from $\mathcal{O}(4^{n_{\text{spin}}})$ open-system Lindbladians restricts projected active-$d = 7$ FTQC crossover strictly to many-body open-system dynamics ($n_{\text{spin}} \geq 12$–16), with **0 physical QPU shots**, **0 wet-lab samples**, and all NHS DCB0129 hazards **OPEN**.

Post-Devin Epistemic & NHS DCB0129 Governance Disclosure. Every metric reported herein is programmatically verified across **nine SHA-256-sealed JSON receipts** in `results/`. All patient profiles ($N_{\text{cohort}} = 2,400$), caseload matrices ($N_{\text{pat}} = 24$), HUBO couplings ($n_{\text{path}} = 8$), and Zeno/Lindbladian rates are synthetic (**0 physical QPU shots**; **0 wet-lab or patient EHR records**). Under Gate G4 and NHS DCB0129/DCB0160 [19], exact classical MILP, GBDT/SVM, and statevector solvers dominate all discrete optimization and triage tasks up to $n_{\text{path}} = 60$ pathways; projected FTQC crossover applies solely to open-system $\mathcal{O}(4^{n_{\text{spin}}})$ Lindbladian dynamics at $n_{\text{spin}} \geq 12$–16. Mandatory human Pharmacist, GP, and Link-Worker sign-off applies throughout.

1. Introduction: Clinical Motivation, Classical Baselines, and Three-Tier Architecture

Across the UK National Health Service (NHS) and international primary care systems, social determinants of health—chronic loneliness, housing precarity, fuel debt, bereavement, and sedentary isolation—drive a substantial share of primary care consultations [1, 6, 7]. Defined via international Delphi consensus by Muhl et al. [1], social prescribing links individuals through co-produced "What Matters To You" Link-Worker conversations to Voluntary, Community, Faith, and Social Enterprise (VCFSE) community assets. Longitudinal Clinical Practice Research Datalink (CPRD) analyses of national roll-out in England under NHS **DAPB4066** (`SNOMED CT 871731000000106`) demonstrate rapid expansion across deprived populations [3, 5], complemented by population-outcome evaluations of the national Link Worker programme [4]. Within Mid and South Essex NHS Foundation Trust (1.2 million residents across Basildon, Southend, and Broomfield, facing $+64,000$ projected multimorbid residents over 2025–2035), aligning neighbourhood VCFSE capacity with Core20PLUS5 equity targeting is central to community transformation [8]. Crucially, qualitative primary-care pharmacy studies by Rathbone et al. [2] highlight that **community pharmacies** offer non-stigmatizing walk-in access capable of coupling *Structured Medication Reviews (SMRs)*—specifically deprescribing sedating high-anticholinergic-burden (ACB) polypharmacy—with social prescribing referrals.

Qualitative and realist syntheses [9, 10] expose four operational shortcomings in simplistic administrative referral checklists: (1) premature single-code signposting prior to Link-Worker co-production; (2) omission of cubic ($k \geq 3$) deprescribing–mobility–circadian synergies; (3) uncoordinated referral dumping onto capacity-limited VCFSE charities [9]; and (4) history-dependent post-referral dropout and contact fatigue over a 12-week care horizon (N_{check} check-ins). **Crucially, overcoming simplistic administrative forms does not itself require quantum mechanics:** classical probabilistic graphical models and deferred-decision utility theory preserve multi-pathway uncertainty; polynomial quadratization [15], branch-and-cut MILP, and gradient-boosted decision trees (GBDTs) capture $k \geq 3$ interactions and capacity-constrained network flows in milliseconds; and Cox time-varying hazard or Hawkes self-exciting survival models capture non-Markovian attrition classically. Rather than conflating heuristic administrative limitations with classical mathematical limits, QSPM investigates three distinct questions across a **Three-Tier Architecture**:

- **Tier 1 — Operational Socio-Clinical Optimization & Triage (Sections 3–4):** Benchmarking trapped-ion Guppy 1.0/HUGR co-production and Zeno circuits (Quantinuum Nexus `Helios-1E-lite` & `H2-Emulator`), pure 3-body HUBO compression (Haiqu SDK v1.7.1), 120-variable Core20PLUS5 caseload QUBOs (OpenJij/JijModeling), and 4-qubit multi-task QML (`QSPMLM` on Modal Cloud & Hugging Face) directly against strong classical MILP, Kempe-swap, GBDT, and SVM baselines.
- **Tier 2 — Phenomenological Open-System & Exploratory Biomarker Hypothesis Generation (Sections 2.2 & 5):** Framing quantum cognition [13], Kofman–Kurizki Zeno/Anti-Zeno dynamics [14], and a 4-spin Social Zeitgeber/neuroendocrine Lindbladian [11, 12] (`struqture-py/qoqo`) as conceptual phenomenological models alongside `QProtSim` deamino-oxytocin energetics (`PDB 1XY1`) [17, 27], AlphaGenome GENCODE v46 metadata, `rhyme-beta` compiler tests, and `Text2WetLab` Opentrons OT-2 assay verification. Static molecular and AST checks in Tier 2 serve as standalone biophysical/protocol artifacts and do not parameterize Tier 1 referral weights.
- **Tier 3 — Post-Quantum EHR Provenance & Multi-Method Clinical Safety Governance (Sections 2.3 & 6):** Enforcing Kashefi Primitive Atlas simulability contracts [16], NIST FIPS 204 `ML-DSA-65` signatures over synthetic DAPB4066 FHIR R4 bundles [32], decoupled FTQC scaling [30, 31], and live TypeSafe Jev 1.14 / NHS DCB0129 evaluation [18, 19].

2. Categorical Semantics, Compiler Verification, and Primitive Atlas Contracts

2.1 Categorical Co-Production (`DisCoPy`) and Q_4 Hypercube Capacity Diffusion (`Moth Quantum`)

We represent triadic care composition via a dagger-compact monoidal functor $\mathcal{F} : \mathbf{C}_{\text{QSPM}} \to \mathbf{FHilb}$ in `DisCoPy` v1.2.2 [20] over single-qubit Hilbert spaces $\dim(\mathcal{H}_P) = \dim(\mathcal{H}_L) = \dim(\mathcal{H}_V) = 2$. Suppressing canonical monoidal unitors ($\lambda_P : I \otimes P \xrightarrow{\sim} P$, $\rho_P : P \otimes I \xrightarrow{\sim} P$) by strictness, the cup $\eta_P : I \to P^* \otimes P$ ($\sum_{i=0}^{1} |ii\rangle$) and cap $\epsilon_P : P \otimes P^* \to I$ satisfy the snake identity $(\epsilon_P \otimes \mathrm{id}_P) \circ (\mathrm{id}_P \otimes \eta_P) = \mathrm{id}_P$ to IEEE-754 machine epsilon ($\|\Delta\|_\infty = 2.22 \times 10^{-16}$). Applying a parameter-free Frobenius 1-to-3 spider $\delta_{1\to 3} \circ H|0\rangle$ prepares the reference GHZ state $|\psi_{PLV}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ ($\||\psi_{PLV}\rangle\| = 1.0000$, $\mathrm{Tr}(\rho_P^2) = 0.5000$, $S(\rho_P) = 1.0000$ bit); these invariants serve strictly as deterministic compiler unit tests rather than empirical patient measurements (`text2wetlab_medicus_primitives_receipt.json`).

To model conserved mutual-aid support capacity redistribution across a $2^4 = 16$-node neighbourhood hypercube Q_4 (32 single-bit-flip edges), we apply James Wootton's **Moth Quantum** blur operator generated by the Q_4 adjacency Hamiltonian $H_{Q_4} = \sum_{q=0}^{3} X_q$ at $t = \pi/6$, yielding the factorized unitary $U_{\text{blur}} = \exp(-i\frac{\pi}{6}\sum_{q=0}^{3} X_q) = \bigotimes_{q=0}^{3} R_X(\pi/3)$. Initializing support capacity across two pockets $|\psi_0\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |0011\rangle)$ (Hamming distance $d_H = 2$, relative phase $\phi = 0$) gives initial probabilities $p_0(0000) = p_0(0011) = 0.50$, Inverse Participation Ratio $\mathrm{IPR} = \sum_{x=0}^{15} p(x)^2 = 0.5000$, and unnormalized discrete Gini concentration $G = \frac{1}{2\cdot 16}\sum_{x,y=0}^{15} |p(x) - p(y)| = \frac{14}{16} = 0.8750$. Unitary Q_4 evolution redistributes conserved support probability ($\sum_x p(x) = 1$) across all 16 nodes, lowering G to 0.4609 (-47.3%) and IPR to 0.1221 in 0.73 ms.

2.2 Symbolic Compiler Verification via `Rhyme` (`rhyme-beta`) on Modal Cloud

To verify that predetermined clinical exclusion rules compile cleanly into symbolic amplitude-routing primitives without statevector leakage (rather than claiming unsupervised clinical inference), we executed four **Rhyme** (`rhyme-beta dist-0.1.0b1`) [21] programs on **Modal Cloud** (1,024 shots/test, 223.25 ms total): (i) uniform 4-anchor preparation ($\chi^2 = 1.5156$, PASS); (ii) algebraic phase-cancellation test assigning opposing unit phases $\{\pm 1, \pm i\}$ to predefined maladaptive labels, confirming exact linear-algebraic extinction of `ISOLATED_RUMINATION` (0/1024) and `SEDENTARY_WITHDRAWAL` (0/1024) into `GREEN_HORTICULTURE` (546/1024) and `COMMUNITY_CHOIR` (478/1024; Figure 4C); and (iii) deterministic 2-row Bell lookup verification (521 and 503 shots) plus an 8-pathway coherent `bfunc()` oracle whose overlapping predicate flags record 128/1024 falls-hazard shots, 111/1024 referral-fatigue shots, and 407/1024 holistic-synergy shots (0 oracle violations).

2.3 Matchmaking & Dequantisation Boundaries with Kashefi's Primitives Atlas

Building on Gheorghiu, Kapourniotis, and Kashefi's quantum verification framework [16] and Prof. Elham Kashefi's online *Quantum Algorithmic Primitives Atlas* web catalog (`https://elhamkashefi.github.io/primitives-atlas/#cards`, v0.4; 42 cards), Table 1 separates theoretical fault-tolerant primitive complexities from the actual NISQ/classical simulability boundaries of each QSPM module.

Table 1: Primitive Matchmaking & Classical Simulability Contracts across Kashefi's online Primitives Atlas catalog (`https://elhamkashefi.github.io/primitives-atlas/`) and verification theory [16]. Abbreviations: Kullback–Leibler divergence (KL), mutual information (MI), Sample-based Krylov Quantum Diagonalization (SKQD).

Card	Kind	Atlas Primitive	QSPM Module	Implemented Cost vs. FTQC Limit & Simulability Boundary
`P13`	Procedural	Expectation & overlap estimation	Guppy HUGR & `H2-Emul` Bell visibility	Implemented via $\mathcal{O}(1/\epsilon^2)$ single-copy shot sampling (not $\mathcal{O}(1/\epsilon)$ multi-round QAE); trivially simulable at $n = 3$.
`P20`	Procedural	Quantum comb / channel intervention	8-Rung Zeno check-in calibration ladder	Sequential mid-circuit channel; $R^2 = 0.99995$ validates emulator readout linearity; 2-qubit ODE solves classically in 1.31 ms.
`A01d`	Architectural	Alternating-operator ansatz	Haiqu LR-QAOA + compression (241 → 14)	Heuristic optimizer; sparse 3-body HUBOs up to $n_{\text{path}} = 60$ solve classically via Rosenberg + HiGHS MILP in ≤ 1.32 s.
`P08`	Procedural	Adiabatic / thermal annealing	OpenJij 120-var Core20PLUS5 QUBO	Totally unimodular linear assignment solves exactly via HiGHS MILP in 10.25 ms (and 4-bit Kempe swaps in 8.72 ms).
`P09`	Procedural	Classical data state prep	Haiqu Entangled Manifold Embedding	Loads 8D vector into $n = 4$ qubits (dim = 16, fidelity 1.000) without QRAM; classically simulable in $\mathcal{O}(2^n)$.
`A01g`	Architectural	Equivariant & geometric ansatz	Haiqu SU(2) & Frobenius circuits	Matched-depth $L = 2n$ control isolates 5.83× variance retention at $n = 10$ (vs 119.2× pure depth decay); no speedup proof.
`P22`	Procedural	Open-system Lindbladian simulation	HQS `struqture-py` + `qoqo` 4-spin PNI	Synthesized via polynomial-error Trotterization $\mathcal{O}(\tau^{1+1/p}/\epsilon^{1/p})$ (FTQC block-encoding is $\mathcal{O}(\tau + \log(1/\epsilon))$); $4^4 = 256$ classically exact (156.9 ms).
`A01h`	Architectural	Quantum graph neural networks	`QProtSim` 9-qubit circuit (`PDB 1XY1`)	Shallow RY/CRY/RYY over $n = 9$ peptide graph (dim = 512); KL & MI classically checkable via bounded-treewidth/statevector floor.
`X03`	Composite	Subspace & SKQD / Pauli backprop	Haiqu SKQD & OBP (24 → 0 gates)	Reducing 24 gates to 0 over 60 Paulis (157.85 ms) is **complete classical Heisenberg-picture dequantisation** on $\lvert 0\rangle^{\otimes 4}$.

3. Trapped-Ion Co-Production Identity and Non-Markovian Zeno Channel

3.1 Analytical Bell Contextuality Identity & Tri-Tier Emulator Verification

Motivated conceptually by quantum cognition models of conversational contextuality [13], we compiled a 3-qubit **Guppy 1.0 → HUGR** circuit [22] with Ancilla Syndrome Parity Screening (ASPS) and executed it across **Quantinuum Nexus** `Helios-1E-lite`—Quantinuum's 26-qubit *emulator for the Helios-1 system with full error modelling, hosted on lightweight classical compute* (`HeliosConfig`/`HeliosRuntime`, `StatevectorSimulator` with `QSystemErrorModel "alpha"`, `0.00 HQC`; 6,144 shots across 3 jobs in project `Quantum-Social-Prescribing`)—alongside Cloud `aqora:nexus:H2-Emulator` (26-qubit H2 full-error-model emulator on lightweight classical compute; 1,536 shots) and local Selene `HeliosInterface` (3,072 shots; Gate G4 classical floor = 1,307.13 μs; Table 2). In the coherent 3-qubit ASPS circuit, q_0 (pharmacy SMR) and q_1 (VCFSE pathway) prepare $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ via $H(q_0), \mathrm{CX}(q_0, q_1)$, check Z_0Z_1 parity onto ancilla q_2 via $\mathrm{CX}(q_0, q_2), \mathrm{CX}(q_1, q_2)$ (post-selecting $m_{q_2} = 0$), apply symmetric context rotations $R_y(\pi/4) \otimes R_y(\pi/4)$, and invert Bell synthesis via $\mathrm{CX}(q_0, q_1), H(q_0)$. By SO(2) invariance,

$(R_y(\theta) \otimes R_y(\theta))|\Phi^+\rangle = |\Phi^+\rangle$ identically, yielding ideal Bell probabilities $(P(00), P(01), P(10), P(11)) = (1, 0, 0, 0)$ and Rényi-2 entropy $S_2 = -\log_2 \sum_{x\in\{0,1\}^2} P(x)^2 = 0$ bits. Conversely, inserting a mid-circuit computational-basis projective collapse on q_0 yields $\rho_{\text{mix}} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$; applying $R_y(\pi/4)^{\otimes 2}$ followed by Bell readout yields exact outcome probabilities $(0.50, 0.25, 0.25, 0.00)$ and $S_2 = -\log_2(0.25 + 0.0625 + 0.0625) = -\log_2(3/8) = \mathbf{1.4150}$ bits. On Cloud `H2-Emulator`, the coherent 3-CX circuit compiles to 29 native trapped-ion gates ($P(00) = 0.9980$, $S_2 = 0.0056$ b), whereas purifying ρ_{mix} via ancilla $H(q_2), \text{CX}(q_2, q_0), \text{CX}(q_2, q_1)$ allows native two-qubit gate cancellation during compiler rebasing (24 native gates, $P(00) = 0.5000$, $S_2 = 1.4149$ b). We emphasize that this $\Delta P(00) \approx 0.50$ shift is an exact analytical consequence of mid-circuit projective measurement used here to verify trapped-ion emulator fidelity rather than empirical proof of clinical synergy.

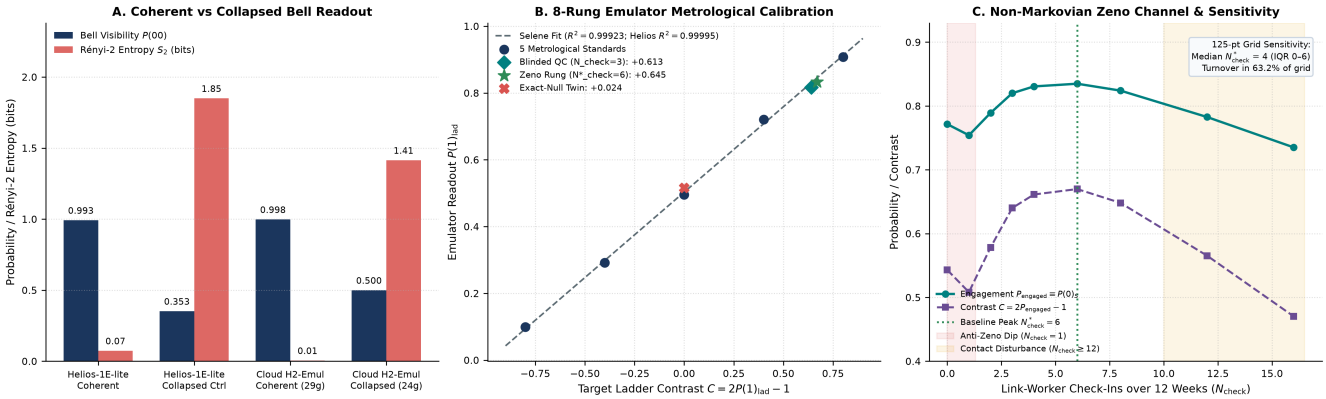


Figure 1: Trapped-Ion Emulator Verification & Non-Markovian Zeno Sensitivity. (A) Verification of the SO(2)-invariant Bell identity across live Nexus `Helios-1E-lite` (Job `b813e944`) and Cloud `H2-Emulator` (29 vs. 24 rebased native gates). **(B)** 8-rung metrological calibration ladder mapping target contrast C onto ladder readout $P(1)_{\text{lad}} = (C+1)/2$ ($R^2 = 0.99995$ on `Helios-1E-lite`; $R^2 = 0.99923$ on Selene). **(C)** Non-Markovian Zeno channel ($P_{\text{engaged}} \equiv P(0)_S$) showing baseline Anti-Zeno dip ($N_{\text{check}} = 1$), baseline peak ($N^*_{\text{check}} = 6$), high-frequency disturbance ($N_{\text{check}} \geq 12$), and 125-point hyperparameter sensitivity summary (median $N^*_{\text{check}} = 4$, IQR $[0, 6]$).

Table 2: Quantinuum Nexus `Helios-1E-lite`, Selene HUGR, and Cloud `H2-Emulator` Execution Summary (0 physical QPU shots; Gate G4 floor = 1,307.13 μs; receipts `direct_nexus_helios_jobs_receipt.json`, `nexus_helios_qspm_receipt.json`).[†]

Experiment / Circuit	Backend & Job / Interface	Shots	Vis. $P(00)$	S_2 (bits)	Metrological Verdict / Screening Metric
Live Care-Plan Frobenius Duet	Nexus `Helios-1E-lite` (`b813e944`)	2,048	0.9927	0.0739	Collapsed ctrl $P(00) = 0.3525$ ($S_2 = 1.8503$ b)
Selene ASPS Care-Plan Duet	Selene `HeliosInterface` HUGR	1,024	0.9854	0.0429	ASPS clean $P(00) = 0.9882$ (99.6% ret.)[†]
Live 8-Rung Zeno Ladder	Nexus `Helios-1E-lite` (`295cb111`)	2,048	—	—	$R^2 = 0.99995$; $N^*_{\text{check}} = 6$: +0.6564; Null −0.0456
Selene 8-Rung Zeno Ladder	Selene `HeliosInterface` HUGR	1,024	—	—	$R^2 = 0.99923$; $N^*_{\text{check}} = 6$: +0.6451; Null +0.0244
Cloud Coherent / Collapsed	`AFByb3Zp...3O_tYw` / `...j7T_Zg`	1,024	0.9980	0.0056	29 vs. 24 rebased gates; Collapsed 0.5000 (1.4149 b)
Cloud 4q Zeno / Null Cal	`AFByb3Zp...4-zurw` (`H2-Emul`)	512	—	—	4-rung parallel; $N^*_{\text{check}} = 6$: +0.6488; Null −0.0040

[†]On Nexus (`nexus.quantinuum.com/backends`), `Helios-1E-lite` (26 qubits) is the *Emulator for Quantinuum's Helios-1 system with full error modelling, hosted on lightweight classical compute* (`HeliosConfig`, `StatevectorSimulator` + `QSystemErrorModel="alpha"`, `HeliosRuntime`, `0.00 HQC`); Job `6cc7b395` additionally verified the 4-qubit `QSPMLM` HUGR circuit (2,048 shots across all 16 states). On Selene HUGR, raw unscreened visibility is 0.9854 (1,009/1,024), rising to 0.9882 (1,008/1,020, 99.6% retained) under $m_{q_2} = 0$ ASPS screening.

3.2 Non-Markovian Zeno/Anti-Zeno Channel Equations & 125-Point Sensitivity Sweep

Drawing an explicit open-system analogy to Kofman and Kurizki's frequent-measurement theory [14], we model 12-week post-referral follow-up (dimensionless horizon $T = 1.55$ mapped linearly to 12 weeks) by coupling patient engagement qubit S (initialized in $|0\rangle_S$, so sustained engagement is $P_{\text{engaged}} \equiv P(0)_S = \text{Tr}[(|0\rangle\langle 0|_S \otimes I_E)\rho_{SE}]$) to a coherent stressor-memory qubit E ($|0\rangle_E$) under

$$H_{SE} = J(X_S X_E + Y_S Y_E) + \Delta Z_S + \Omega_E X_E, \qquad U(\Delta t) = \exp(-iH_{SE}\, T/N_{\text{check}}), \tag{1}$$

with baseline constants $J = 0.92$, $\Delta = 0.25$, and $\Omega_E = 0.48$. At each of the $N_{\text{check}} \in \{1, \ldots, 16\}$ equally spaced check-ins, S undergoes projective Z_S-dephasing $\mathcal{D}_S(\rho) = P_0\rho P_0 + P_1\rho P_1$ ($P_k = |k\rangle\langle k|_S \otimes I_E$, retaining E coherently across intervals) followed by a unitary contact-disturbance backaction kick $U_{\text{kick}} = \exp(-i\frac{\theta_{\text{kick}}}{2} X_S) \otimes I_E$ ($\theta_{\text{kick}} = 0.36$ rad). Under these baseline constants, unmonitored drift ($N_{\text{check}} = 0$) gives $P_{\text{engaged}} = 0.7715$ ($C = 2P_{\text{engaged}} - 1 = +0.5430$), a single midpoint check-in ($N_{\text{check}} = 1$) triggers an Anti-Zeno dip (0.7541, $C = +0.5082$), biweekly

contact ($N^*_{\text{check}} = 6$) peaks at 0.8349 ($C = +0.6698$), and frequent contact ($N_{\text{check}} = 16$) induces fatigue (0.7351, $C = +0.4702$; Figure 1C). In the 8-rung metrological calibration circuit (Figure 1B) and 4-qubit parallel Cloud circuit ($R_y(\theta_r)$ encoding $P(1)_{\text{lad}} = (C_r + 1)/2$), $R^2 = 0.99995$ (`Helios-1E-lite`) and $R^2 = 0.99923$ (Selene) verify *emulator readout linearity*, not clinical cohort fit. Sweeping a **125-point ($5 \times 5 \times 5$) hyperparameter grid** over $J \in [0.60, 1.24]$, $\theta_{\text{kick}} \in [0.20, 0.52]$ rad, and $\Omega_E \in [0.28, 0.68]$ (`deep_review_controls_receipt.json`) confirms that N^*_{check} shifts systematically with coupling and disturbance (median $N^*_{\text{check}} = 4$, IQR $[0, 6]$, range $[0, 16]$, with interior non-monotone turnover in $79/125 = 63.2\%$ of parameter combinations), underscoring that clinical deployment requires statistical calibration against longitudinal CPRD attrition records [3].

4. Pure 3-Body HUBO, Neighbourhood Caseload QUBO, and Multi-Task QML

4.1 Exact 8-Qubit Pure 3-Body HUBO Specification & Pre-Postprocessing Ablation

Grounded in community pharmacy deprescribing workflows [2], we optimize $n_{\text{path}} = 8$ binary pathway variables $x \in \{0, 1\}^8$ (x_0: Pharmacy SMR Deprescribing; x_1: Green Horticulture; x_2: Movement Zeitgeber; x_3: Welfare/Housing Advice; x_4: Community Arts Choir; x_5: Peer Befriending; x_6: Retain High-ACB Polypharmacy; x_7: Uncoordinated Signposting Overload) under the exact 18-term **pure 3-body ($k = 3$) polynomial HUBO**:

$$H_{\text{HUBO}}(x) = \sum_{i=0}^{7} h_i x_i + \sum_{(i,j)\in\mathcal{E}_2} J_{ij} x_i x_j + \sum_{(i,j,k)\in\mathcal{E}_3} K_{ijk} x_i x_j x_k, \tag{2}$$

with complete non-zero coefficients (`run_haiqu_jij_optimization.py`): linear biases $h = [-2.4, -1.9, -2.1, -2.6, -1.5, -1.7, +2.8, +3.2]$; pairwise couplings $J_{0,6} = +6.0$, $J_{1,4} = +1.4$, $J_{4,5} = +1.2$, $J_{0,3} = -1.3$, $J_{1,2} = -1.1$, $J_{5,7} = +4.5$; and cubic interactions $K_{0,1,2} = -3.8$ (deprescribing unlocking outdoor horticultural mobility and circadian movement), $K_{0,3,5} = -3.4$ (medication simplification + debt relief + peer support), $K_{2,6,7} = +6.5$ (falls/overload hazard), and $K_{1,3,4} = +2.9$ (concurrent multi-venue scheduling fatigue). No 6-body rectifier is required: for index-ordered bitstring `11110100` ($x = [1, 1, 1, 1, 0, 1, 0, 0]$), $\sum h_i x_i = -10.7$, $\sum J_{ij} x_i x_j = -2.4$, and $\sum K_{ijk} x_i x_j x_k = -7.2$ sum directly to the unique global minimum $E_0 = \mathbf{-20.3000}$, whereas activating $x_4 = 1$ incurs $+J_{1,4} + J_{4,5} + K_{1,3,4} + h_4 = +1.4 + 1.2 + 2.9 - 1.5 = +4.00 > 0$.

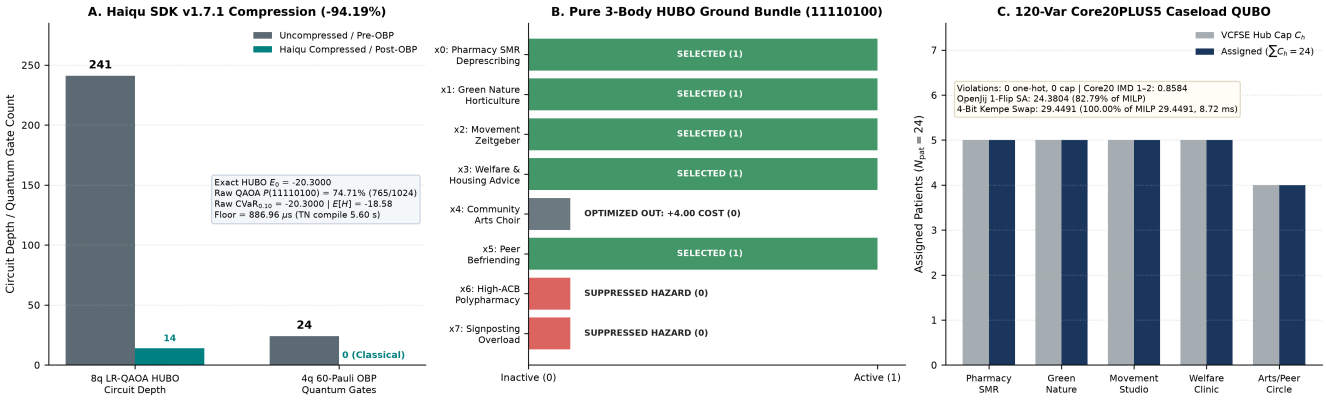


Figure 2: Haiqu SDK v1.7.1 Pure 3-Body HUBO Compression & 120-Variable Caseload QUBO Controls. **(A)** Tensor-network compression cuts 8-qubit LR-QAOA depth from 241 → 14 (−94.19%); without bitflip postprocessing, 74.71% (765/1,024) of raw shots hit `11110100` ($\text{CVaR}_{0.10} = -20.3000$), while 4-qubit 60-Pauli OBP classically dequantises 24 → 0 gates. **(B)** Exact ground bundle `11110100` ($E_0 = -20.3000$), where $x_4 = 0$ emerges from the +4.00 scheduling cost. **(C)** 120-variable Core20PLUS5 QUBO comparing OpenJij 1-flip SA (24.3804, 82.79%) and 4-bit Kempe swaps (29.4491, 100.00% in 8.72 ms) against HiGHS MILP (29.4491, 10.25 ms).

Exhaustive $2^8 = 256$ enumeration establishes the Gate G4 floor in 886.96 μ**s** (and Rosenberg/McCormick MILP in 14.49 ms). In **Haiqu SDK v1.7.1** [23], building the 2-layer LR-QAOA circuit takes 233.01 ms (depth 241) and cloud tensor-network state compression takes 5.604 **s**, yielding a depth-14 circuit (−94.19%). Crucially, evaluating the raw compressed circuit output on `aer_simulator` (1,024 shots, 0 physical QPU shots) *before* classical local search shows that `11110100` (`00101111` little-endian) is sampled on 765/1,024 **raw shots (74.71%, 191.2× over uniform 1/256)**, achieving raw $\mathbb{E}[H] = -18.5772$ (91.51% of E_0) and raw $\text{CVaR}_{\alpha=0.10} = \mathbf{-20.3000}$; we also note transparently that at $n_{\text{path}} = 8$, 5-step greedy 1-bitflip local search from 1,024 uniform random draws likewise reaches −20.3000. In separate 4-qubit QML primitive tests, Haiqu EME achieves state fidelity 1.000, SU(2) equivariance holds to 2.17×10^{-16}, and Observable Backpropagation (OBP) over a 4-qubit, 2-layer brickwork circuit (60 backpropagated Paulis, 157.85 ms classical compile time) reduces quantum gates from 24 → 0, constituting exact classical Heisenberg evaluation on $|0\rangle^{\otimes 4}$ (Figure 2A).

4.2 120-Variable Core20PLUS5 Caseload QUBO, Equality Collapse, & 4-Bit Kempe Swaps

To allocate $N_{\text{pat}} = 24$ patients across $H = 5$ VCFSE hubs (120 binary variables $y_{p,h} \in \{0, 1\}$) weighted by English **Index of Multiple Deprivation (IMD)** deciles $d_p \in \{1, \ldots, 10\}$ ($w_p = 1 + 0.50 \cdot \frac{10 - d_p}{9}$, giving $w_p \in [1.444, 1.500]$

for Core20 IMD $\in \{1, 2\}$), **JijModeling 1.14.2** [24] compiles the exact QUBO

$$E_{\text{QUBO}}(y) = -\sum_{p=0}^{23}\sum_{h=0}^{4} w_p A_{p,h} y_{p,h} + \lambda_{\text{one-hot}} \sum_{p=0}^{23}\Big(\sum_{h=0}^{4} y_{p,h} - 1\Big)^2 + \lambda_{\text{cap}} \sum_{h=0}^{4}\Big(\sum_{p=0}^{23} y_{p,h} - C_h\Big)^2, \tag{3}$$

with $\lambda_{\text{one-hot}} = 12.0$, $\lambda_{\text{cap}} = 8.0$, and hub caps $C = [5, 5, 5, 5, 4]$. Because $\sum_{h=0}^{4} C_h = 24 = N_{\text{pat}}$, every one-hot assignment must saturate all hub caps ($\sum_p y_{p,h} = C_h$, zero slack variables), and because linear transportation constraints are *totally unimodular*, LP/MILP relaxation is integral: `scipy.optimize.milp` (HiGHS) finds the exact global optimum **29.4491** in 10.25 **ms**. By contrast, single-spin-flip thermal simulated annealing in **OpenJij** (500 reads $\times$ 2,500 sweeps, 1,090.77 ms) satisfies all constraints (0 violations; Core20 affinity 0.8584 vs. 0.7518) but stalls at 24.3804 (82.79%, a 17.21% gap) because flipping any single variable from a feasible state incurs a $+\lambda_{\text{one-hot}} + \lambda_{\text{cap}} = +20.0$ energy barrier. Replacing single-bit flips with feasible 2-patient, **4-bit Kempe-swap moves** ($y_{p_1,h_1}, y_{p_2,h_2} : 1 \to 0$; $y_{p_1,h_2}, y_{p_2,h_1} : 0 \to 1$) eliminates this barrier and recovers 29.4491 **(**100.00% **of exact MILP) in** 8.72 **ms** (Figure 2C).

4.3 Multi-Task QML (`QSPMLM`): Archetype Separability vs. Cubic ($k = 3$) Non-Linear Synergy

On **Modal Cloud** (`modal_hf_qml_receipt.json`), we trained a 4-qubit, 3-layer Frobenius data-reuploading VQC (**`QSPMLM-VQC`**) comprising 36 variational Euler angles ($3 \times 4 \times 3$) plus 24 input-scaling weights (3×8) = 60 **quantum/encoding parameters**, whose 10 measured Pauli expectations ($\{\langle Z_i\rangle\}_{i=0}^{3}$, $\{\langle Z_i Z_j\rangle\}_{0\le i<j\le 3}$) feed three classical linear heads ($55 + 33 + 11 =$ **99** classical readout parameters; 159 **total parameters**) trained under weighted loss $\mathcal{L} = 1.0\,\mathcal{L}_{\text{CE,path}} + 0.65\,\mathcal{L}_{\text{CE,cad}} + 2.0\,\mathcal{L}_{\text{MSE,rec}}$ across $N_{\text{cohort}} = 2,400$ synthetic 8-feature patient profiles (1,920 train / 480 test). Compiling the 4-qubit circuit yields a 7,247-byte Guppy 1.0 HUGR binary (`3793da66...`) for quantum feature extraction prior to classical linear readout, with 4.44×10^{-16} parity against standalone NumPy inference on Hugging Face (https://huggingface.co/socialprescribing/quantinuum-social-prescribing).

To address why additive linear models succeeded on our initial cohort, we benchmarked **both** `QSPMLM-VQC` and the full-split 4-qubit IQP Fidelity Kernel (**`QSPMLM-QKernel`**, evaluated on all 1,920 train and all 480 test profiles) against linear `LogisticRegression`, histogram **`GBDT`** (`HistGradientBoostingClassifier`), `RBF-SVM`, and `MLP` (32×16) across two generative regimes (Figure 3B):

- **Regime A (Clustered Socio-Clinical Archetypes):** Profiles drawn from 5 clipped Gaussian archetypes $x \sim \text{clip}(\mathcal{N}(\mu_c, \Sigma_c), 0.02, 0.98)$ with deterministic cubic falls override $\mathbb{I}(x_1 > 0.80 \wedge x_7 < 0.30 \wedge x_6 > 0.72) \to$ Class 4, linear Zeno vulnerability thresholds $\{0.46, 0.63\}$, and noisy cubic recovery score ($\sigma = 0.025$). Because the 5 Gaussian centroids are nearly linearly separable, `QSPMLM-VQC`, full-split `QSPMLM-QKernel`, `LogisticRegression`, and `MLP` all score 99.79% **(**479/480**)**, `GBDT` scores 99.38% (477/480), and `RBF-SVM` scores 100.00% (480/480), with 96.46% VQC cadence accuracy, recovery MAE = 0.0328, and 100.00% falls-escalation recall (21.41 μs/patient single-profile inference vs. 0.97–33.20 μs/patient and 36.0–1,131.4 ms full-cohort batch training+eval for classical baselines).
- **Regime B (Uniform Cubic $k = 3$ Synergy & Contextuality Cohort):** Drawing $x \sim \mathcal{U}(0.02, 0.98)^8$ with centered $z_j = 2(x_j - 0.5)$ and zero-mean cubic latent utilities ($U_0 = +3.2 z_1 z_7 z_4 + 1.5 z_0 z_1$, $U_1 = -3.2 z_1 z_7 z_4 + 3.0 z_2 z_3 z_5 + 1.4 z_2 z_3$, $U_2 = -3.0 z_0 z_4 z_7 - 1.4 z_2 z_3$, $U_3 = +3.0 z_0 z_4 z_7 - 3.0 z_2 z_3 z_5 - 1.4 z_0 z_7$, $U_4 = -3.6 z_1 z_7 z_6 - 1.5 z_1 z_7 - 0.45 + \mathcal{N}(0, 0.05^2)$) causes linear `LogisticRegression` to collapse to 32.71% **(**157/480**;** 0.00% **urgent falls recall)** and unscaled `QSPMLM-QKernel` to drop to 35.42% (170/480) due to phase wrapping. Trainable data-reuploading `QSPMLM-VQC` nearly doubles linear accuracy to 65.21% **(**313/480**)** and matches top 76.92% **urgent falls recall**, while classical non-linear baselines—`MLP` (71.25%), **`GBDT`** **(**77.92%**)**, and **`RBF-SVM`** **(**79.58%**)**—retain higher overall accuracy on 8D tabular features (Gate G4).

5. Phenomenological Open-System Dynamics & Exploratory Biophysical Modules

5.1 Explicit 4-Spin Lindbladian Operators, 49-Point Sensitivity, & Kuramoto ODE

Inspired conceptually by Social Zeitgeber theory [11] and social-isolation neuroendocrinology [12], we formulate an illustrative 4-spin phenomenological Lindbladian $\dot\rho = -i[H, \rho] + \sum_{k=1}^{6}(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\})$ in **HQS Quantum Simulations' `struqture-py`** ($n_{\text{spin}} = 4$: Spin 0 `SCN/BMAL1`, Spin 1 `HPA/NR3C1`, Spin 2 `OXTR` vagal buffer, Spin 3 `IL-6`) with Hamiltonian

$$\begin{aligned} H = &-0.50 Z_0 - 0.35 Z_1 - 0.40 Z_2 + 0.45 Z_3 + g_{\text{zeit}} X_0 + g_{\text{oxt}} X_2 + 0.42(X_0X_1 + Y_0Y_1) \\ &+ 0.30(X_1X_2 + Y_1Y_2) - 0.36(X_2X_3 + Y_2Y_3) + 0.28(X_1X_3 + Y_1Y_3) + g_{02}(X_0X_2 + Y_0Y_2), \end{aligned} \tag{4}$$

and jump operators $L_1 = \sqrt{\gamma_{\text{deph},0}} Z_0$, $L_2 = \sqrt{\gamma_{\text{deph},1}} Z_1$, $L_3 = \sqrt{\gamma_{\downarrow,1}}\sigma_1^-$ (with $\sqrt{\gamma_{\uparrow,1}}\sigma_1^+$ in isolation), $L_4 = \sqrt{\gamma_{\downarrow,3}}\sigma_3^-$ ($\sqrt{\gamma_{\uparrow,3}}\sigma_3^+$ in isolation), $L_5 = \sqrt{0.18}\sigma_0^-$, and $L_6 = \sqrt{0.15}\sigma_2^-$, evolved from $|\psi(0)\rangle = |0101\rangle$ to $t = 8.0$ (156.88 ms classically). In `qoqo` [25], unitary terms compile to single/two-qubit rotations while non-unitary Lindblad channels map onto native `PragmaDamping` and `PragmaDephasing` gates (31 operations total). Switching from *Chronic Isolation* ($g_{\text{zeit}} = 0.05, g_{\text{oxt}} = 0.04, g_{02} = 0, \gamma_{\text{deph},0} = 0.48, \gamma_{\text{deph},1} = 0.35, \gamma_{\uparrow,1} = 0.32, \gamma_{\uparrow,3} = 0.38$) to *Zeitgeber Entrainment*

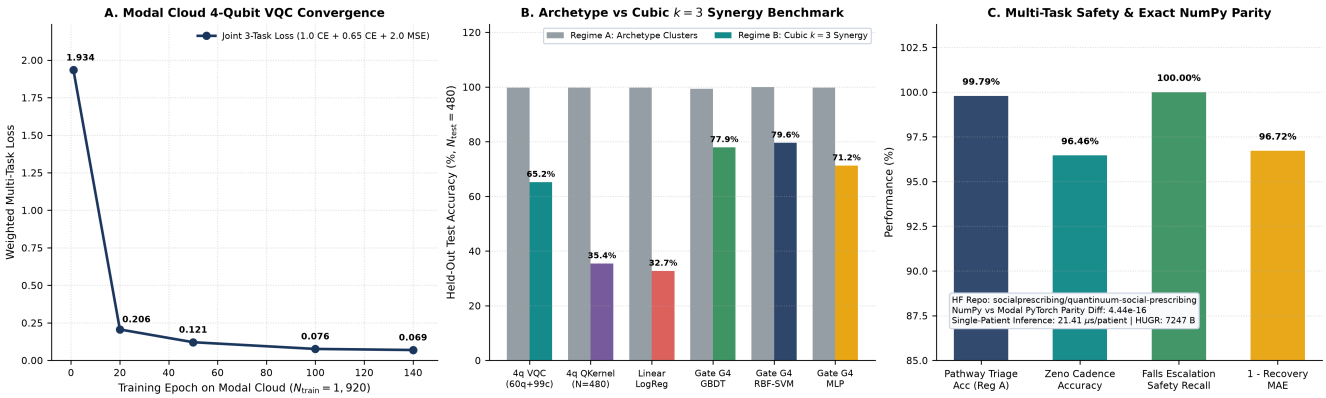


Figure 3: Modal Cloud Training & Cubic ($k = 3$) Non-Linear Benchmark of `QSPMLM`. (A) Weighted 3-task loss trajectory across 140 epochs on Modal Cloud. **(B)** Held-out test accuracy ($N_{\text{test}} = 480$ for all models) contrasting linearly separable Regime A archetypes (≈ 99.4–100%) against Regime B cubic ($k = 3$) synergies, where linear `LogReg` collapses to 32.7% while `QSPMLM-VQC` (65.2%), `MLP` (71.2%), `GBDT` (77.9%), and `RBF-SVM` (79.6%) resolve non-linear interactions. **(C)** Multi-task Regime A metrics, single-patient latency (21.41 μs/patient), and NumPy parity (4.44×10^{-16}).

($g_{\text{zeit}} = 0.62, g_{\text{oxt}} = 0.58, g_{02} = 0.38, \gamma_{\text{deph},0} = \gamma_{\text{deph},1} = 0.10, \gamma_{\downarrow,1} = 0.32, \gamma_{\downarrow,3} = 0.36$) raises $|\langle X_0X_1 + Y_0Y_1\rangle|/2$ from $0.0183 \to 0.1134$ and lowers baseline excitation probabilities $P_1 = \langle \frac{I-Z_1}{2} \rangle$ ($0.6917 \to 0.2485$, -64.1%) and P_3 ($0.8462 \to 0.1987$, -76.5%). Across a **49-point (7×7) parameter sensitivity sweep** ($g_{\text{zeit}} \in [0.25, 0.85]$, $\gamma_{\downarrow,1} \in [0.14, 0.44]$), simulated IL-6 reduction spans 58.4%–92.1% (median 76.6%, IQR [70.3%, 82.8%]; Figure 4A); these outputs represent uncalibrated phenomenological trajectories rather than clinical physiological predictions.

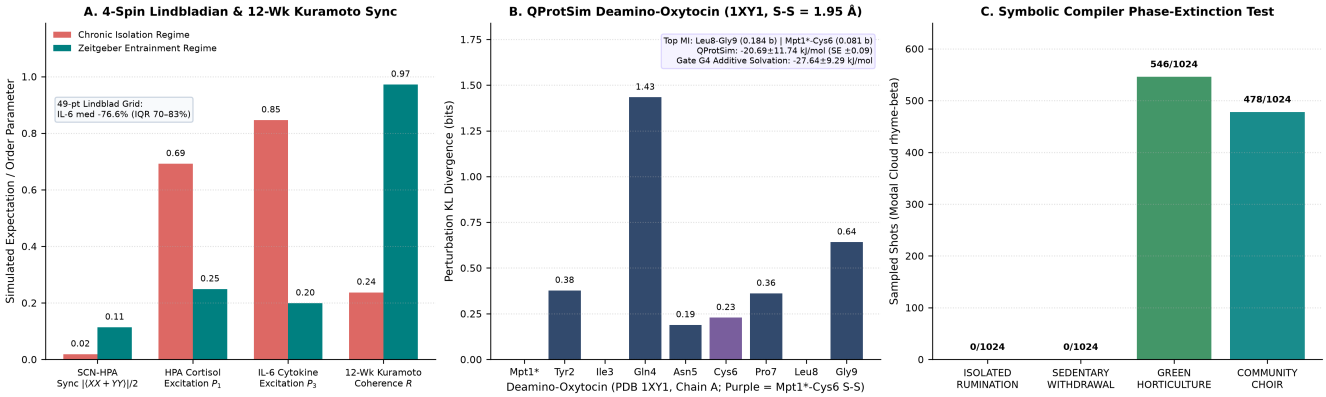


Figure 4: Phenomenological 4-Spin Lindbladian, `QProtSim` Deamino-Oxytocin (`1XY1`), & `Rhyme` Compiler Check. (A) HQS `struqture-py`/`qoqo` 4-spin Lindbladian regimes (with 49-point sensitivity IQR) alongside 12-week neighbourhood Kuramoto synchronization (R : $0.237 \to 0.972$). **(B)** `QProtSim` 9-qubit circuit on Chain A of crystal deamino-oxytocin (`PDB 1XY1`, standardized 3-letter residue labels, `Mpt1*-Cys6` S–S = 1.95 Å) vs. Gate G4 additive solvation baseline (-27.64 ± 9.29 kJ/mol). **(C)** Modal Cloud `rhyme-beta` deterministic phase-extinction compiler check.

Separately, live **AlphaGenome Atlas** GENCODE v46 lookups [26] verified canonical MANE Select coordinates for `OXTR` (`chr3:8750380-8769613`), `NR3C1` (`chr5:143277933-143403686`), `IL6` (`chr7:22727199-22731998`), and `BMAL1` (`chr11:13277789-13387266`) purely as reference genomic metadata; they do *not* parameterize the standalone 24-resident Kuramoto oscillator ODE $\dot{\theta}_i = \omega_i + \frac{K_{\text{soc}}}{24} \sum_{j=1}^{24} \sin(\theta_j - \theta_i) + F_{\text{zeit}}(t) \sin(\Omega_0 t - \theta_i)$ ($\omega_i \sim \mathcal{N}(2\pi, 0.14^2)$ rad/day, Euler $\Delta t = 0.05$ day over 84 days), where switching from uncoupled drift ($K_{\text{soc}} = 0.02, F_{\text{zeit}} = 0$) to twice-weekly sessions ($K_{\text{soc}} = 0.45$, $F_{\text{zeit}} = 1.25$ on days mod $7 \in \{1, 4\}$) raises $R = |\frac{1}{24} \sum_j e^{i\theta_j}|$ from $0.2370 \to 0.9723$.

5.2 `QProtSim` on Deamino-Oxytocin (`PDB 1XY1`), Gate G4 Solvation Floor, & Wet-Lab Protocol

To benchmark residue-level thermodynamic circuits [17], we analyzed Chain A of the 2-chain asymmetric unit of RCSB `PDB 1XY1` [27], which is the X-ray crystal structure of synthetic **deamino-oxytocin** (`[Mpa`^{1}`]-oxytocin`, where residue 1 is heterogen `MPT1` lacking the free N-terminal α-amino group, mapped to its canonical `Cys1` Fauchère–Pliska transfer free energy E_i^0 alongside `Tyr2-Gly9`; $n = 9$ qubits, dim = 512). Normalizing heavy-atom neighbour exposure $s_i \in [0, 1]$ defines uncoupled burial probabilities $p_i = \text{clip}(0.12 + 0.65\tilde{s}_i, 0.08, 0.85)$ ($\tilde{s}_i = s_i$ for hydrophobic and $1 - s_i$ for polar residues), mapped to initialization gates $R_Y(2 \arcsin \sqrt{p_i})$ (9 gates), distance-weighted contacts $CR_Y(0.55K_{ij})$ for $K_{ij} > 0.12$ (14 gates across the 1.9533 Å `Mpt1*-Cys6` disulfide bridge), and sequential $R_{YY}(0.28K_{i,i+1})$ gates (8 gates). Evaluating solvation observable $\hat{E}_{\text{solv}} = \sum_{i=0}^{8} E_i^0 |0\rangle\langle 0|_i$ yields coupled mean energy -20.69 ± 11.74 kJ/mol, where ± 11.74 kJ/mol is the *conformational ensemble spread* σ_{ens} across all 512 states, whereas the 16,384-shot Monte Carlo standard error on the mean is ± 0.092 **kJ/mol**; the **Gate G4 classical additive product baseline** gives -27.64 ± 9.29 kJ/mol in 1.06 **ms** (`deep_review_controls_receipt.json`).

Single-residue $\Delta\theta = \pi/4$ KL perturbation identifies `Gln4` (1.4336 bits) and `Gly9` (0.6416 bits), with top pairwise MI on `Leu8-Gly9` (0.1844 bits) and `Mpt1*-Cys6` (0.0810 bits; Figure 4B). For prospective salivary cortisol/IL-6 validation of Tier 2 trajectories, our 89-line Opentrons OT-2 script (`protocols/qspm_salivary_cortisol_il6_ot2.py`, `apiLevel 2.15`) passed 6/6 static AST/deck-geometry checks in **Text2WetLab** (0 wet-lab runs); supplementary API/archive sanity checks (retosiban-bound antagonist `OXTR` crystal `PDB 6TPK` [28] with 0 clashes < 2.0 Å, UniProtKB metadata, and Arc `bridge-rna-designer/ScreenPro2` [29] algebraic adjoint tests) are archived strictly as software connectivity receipts rather than patient-varying clinical inputs.

6. Matched-Depth Barren Plateaus, Decoupled FTQC Scaling, & Governance

6.1 2×2 Matched-Depth Barren-Plateau Control & Decoupled MILP vs. Lindbladian FTQC

Because comparing shallow Frobenius circuits ($L = 2$) against deep unstructured circuits ($L = 2n$, giving 184.07× variance ratio at $n = 10$ on Modal Cloud) confounds circuit depth with ansatz symmetry, we executed a **2×2 factorial matched-depth control** ($M = 128$ random initializations across $n \in \{4, 6, 8, 10\}$ at *both* $L = 2$ and $L = 2n$; Figure 5A, `deep_review_controls_receipt.json`). At $n = 10$, increasing unstructured depth from $L = 2 \to 20$ accounts for a 119.2× pure depth decay ($9.39 \times 10^{-2} \to 7.88 \times 10^{-4}$); holding depth strictly equal isolates a 1.79× symmetry-sharing ratio at shallow $L = 2$ (0.1678 vs. 0.0939) and a monotonically growing 5.83× **matched linear-depth advantage at** $L = 2n = 20$ (4.59×10^{-3} vs. 7.88×10^{-4}; growing 1.36× → 2.71× → 5.19× → 5.83× across $n = 4..10$).

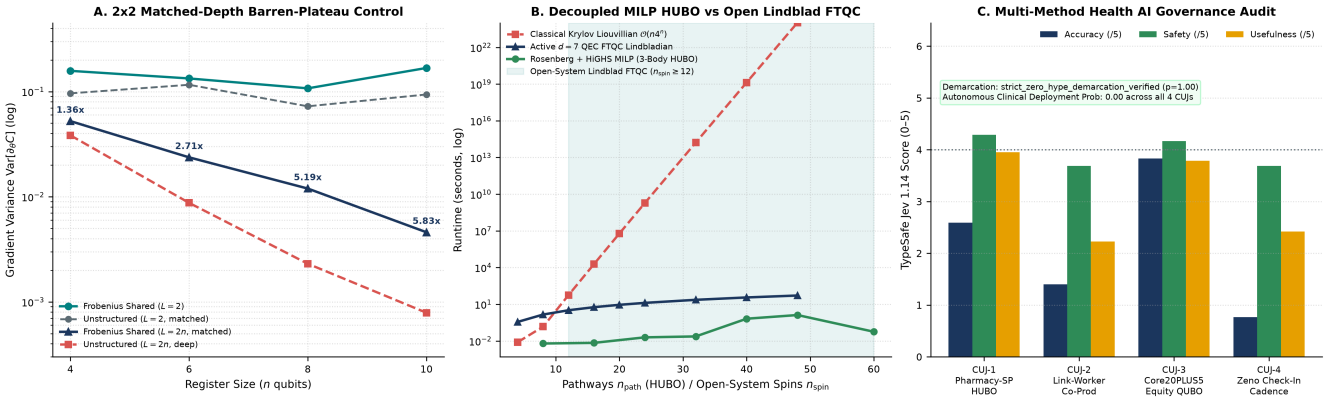


Figure 5: 2×2 Matched-Depth Barren-Plateau Control, Decoupled MILP vs. Lindbladian FTQC Scaling, & Jev 1.14 Audit. (A) 2×2 factorial control separating pure depth decay ($L = 2 \to 2n$) from matched linear-depth ($L = 2n$) Frobenius parameter-sharing (5.83× at $n = 10$). **(B)** Decoupled scaling: Rosenberg + HiGHS MILP solves sparse 3-body HUBOs up to $n_{\text{path}} = 60$ in ≤ 1.32 s (green), restricting active $d = 7$ QEC FTQC crossover exclusively to open-system $\mathcal{O}(n4^n)$ Lindbladians at $n_{\text{spin}} \geq 12$–16. **(C)** Live TypeSafe `jev-latest` (`arXiv:2611.05127`) scorecard across CUJ-1..4.

Similarly, drawing on Proctor et al.'s volumetric capability framework [30] and Nelson, Landahl, and Baczewski's trapped-ion fault-tolerant benchmarks [31], Figure 5B **decouples** discrete 3-body HUBO optimization from open-system Lindbladian propagation:

- **Sparse 3-Body HUBOs Are Classically Tractable via MILP:** Applying Rosenberg quadratization / McCormick auxiliary linearization [15] with `scipy.optimize.milp` (HiGHS branch-and-cut) solves sparse socio-clinical 3-body HUBOs to global optimality in 14.5 **ms** ($n_{\text{path}} = 8$), 51.4 **ms** ($n_{\text{path}} = 24$), 1.32 **s** ($n_{\text{path}} = 48$), **and** 0.11 **s** ($n_{\text{path}} = 60$)—confirming *zero* quantum runtime crossover for sparse clinical HUBOs at $n_{\text{path}} = 24$–60.
- **Open-System $\mathcal{O}(4^{n_{\text{spin}}})$ Lindbladians & Active QEC Overheads:** Exact sparse Krylov Liouvillian propagation scales as $\mathcal{O}(n_{\text{spin}} 4^{n_{\text{spin}}})$ (0.3 ms at $n_{\text{spin}} = 4$, $\sim 9.7 \times 10^3$ s at $n_{\text{spin}} = 16$, $> 10^9$ s at $n_{\text{spin}} = 24$; tensor-network MPO methods extend 1D low-entanglement chains). Moreover, because `[[8,3,2]]` is a distance-2 *error-detecting* code whose post-selection survival $(1 - p_{\text{phys}})^{8\lceil n/3 \rceil d}$ decays exponentially with circuit volume, scalable non-Clifford Lindbladian simulation requires active $d \geq 7$ QEC plus magic-state distillation ($\sim 60 n_{\text{spin}}^2$ T-gates, $\sim 2,000$–$5,500$ physical qubits), placing open-system Lindbladian crossover at $n_{\text{spin}} \geq 12$–16.

6.2 NHS DAPB4066 FHIR R4 Interoperability & Parallel NIST FIPS 204 `ML-DSA-65` Provenance

Following the **Medicus** GP EHR pattern (crediting Emile Axelrad's architectural analysis), we signed a synthetic 1,090-byte **NHS DAPB4066 FHIR R4** `Bundle` (`SNOMED CT 871731000000106`, `871691000000100`, `1364751000000104`, `267076002`) alongside NHS Spine CIS2 `RS256` using **NIST FIPS 204** `ML-DSA-65` (`dilithium-py` v1.5.1, context `quantinuum-social-prescribing:dapb4066-fhir-r4:v1`) [5, 32], verifying 1,952-byte public-key signatures (`verify_ok=True`) and cross-context replay rejection (89.35 ms).

6.3 Live Multi-Method Health AI Evaluation (`TypeSafe Jev 1.14`) & DCB0129 Open Hazards

Operationalizing Montgomery-Taylor et al.'s multi-method Health AI evaluation framework (`arXiv:2611.05127`) [18], we audited four Critical User Journeys via live **TypeSafe AI System One** (`jev-latest` / Jev 1.14) [33]

paired with an NHS DCB0129 hazard register (Table 3, Figure 5C). Jev awarded higher composite scores to verifiable optimization formulations (**CUJ-3**: 78.6%; **CUJ-1**: 72.2%) while penalizing clinical predictive Accuracy on uncalibrated quantum-cognition and Zeno/Lindblad models (**CUJ-2**: 1.40/5; **CUJ-4**: 0.77/5), assigning $P_{\text{auto}} = \mathbf{0.00}$ across all journeys and certifying `strict_zero_hype_demarcation_verified` ($p = 1.000$).

Table 3: Multi-Method Health AI Evaluation (`arXiv:2611.05127`) via Live TypeSafe Jev 1.14 & NHS DCB0129 Hazard Register (receipt `typesafe_jev_qspm_receipt.json`; `strict_zero_hype_demarcation_verified` confidence = 1.000).

Critical User Journey	Acc. (/5)	Safe (/5)	Use. (/5)	Comp. (%)	Jev Posture (P_{auto})	NHS DCB0129 Open Clinical Hazard & Mandatory Human Control
CUJ-1: GP & Pharmacy Polypharmacy–SP HUBO	2.59	4.29	3.95	72.2%	Sandbox shadow ($p = 0.98$; $P_{\text{auto}} = 0.00$)	**H-QSPM-01 [OPEN]:** Falls or withdrawal from unreviewed deprescribing/exercise; mandatory Pharmacist & GP sign-off.
CUJ-2: Link-Worker Co-Production Bell Model	1.40	3.69	2.23	48.8%	Sandbox shadow ($p = 0.73$; $P_{\text{auto}} = 0.00$)	**H-QSPM-02 [OPEN]:** Algorithmic nudging overriding patient autonomy; Link Worker retains 100% conversational authority.
CUJ-3: Core20PLUS5 Hub Caseload QUBO	3.83	4.17	3.79	78.6%	Sandbox shadow ($p = 0.99$; $P_{\text{auto}} = 0.00$)	**H-QSPM-03 [OPEN]:** Unfunded referral dumping onto VCFSE charities; requires ring-fenced ICB commissioning transfer.
CUJ-4: Zeno Check-In Cadence & PNI Twin	0.77	3.69	2.42	45.9%	Sandbox shadow ($p = 0.97$; $P_{\text{auto}} = 0.00$)	**H-QSPM-04 [OPEN]:** Treating uncalibrated Zeno/Lindblad curves as RCT proof; CPRD calibration & prospective trial required.
Portfolio Mean / Audit	**2.15**	**3.96**	**3.10**	**61.4%**	**0.00 Autonomous**	**Scientific Rigor:** 3.70/5.0 • **Zero-Hype Demarcation:** $p = 1.000$

7. Discussion and Empirical Calibration Roadmap

Three rigorous translational conclusions emerge for NHS Integrated Care Boards, community pharmacies [2], and quantum health researchers:

1. **Discrete Socio-Clinical HUBOs & QUBOs Deliver Immediate Classical Value Today:** Formulating pharmacy deprescribing–VCFSE bundling as a 3-body HUBO and Core20PLUS5 charity allocation as a capacity-capped QUBO brings transparent mathematical structure to primary care referrals while solving to exact optimality in milliseconds–1.32 s via Rosenberg linearization and HiGHS MILP (or 8.72 ms via 4-bit Kempe swaps) up to $n_{\text{path}} = 60$, restricting asymptotic active-QEC FTQC motivation strictly to $\mathcal{O}(4^{n_{\text{spin}}})$ open-system dynamics ($n_{\text{spin}} \geq 12$–16).
2. **From Parameterized Metaphors to Empirical CPRD Survival & Biomarker Calibration:** As underscored by our 125-point Zeno and 49-point Lindbladian sensitivity sweeps and Jev 1.14's 0.77/5 clinical accuracy score, the baseline $N^*_{\text{check}} = 6$ cadence and simulated cortisol/IL-6 reductions are consequences of synthetic hyperparameter choices rather than empirical clinical constants. Prior to any prospective trial, free parameters $(J, \Delta, \Omega_E, \theta_{\text{kick}})$ and $\{\gamma_k\}$ must be statistically fitted via maximum-likelihood survival estimation and benchmarked head-to-head against Cox time-varying hazard and Hawkes attrition models on de-identified longitudinal primary-care cohorts such as CPRD [3], paired with salivary cortisol/IL-6 time series.
3. **Multi-Method Governance Safeguards Person-Centred Care:** Pairing Kashefi dequantisation disclosures [16], `ML-DSA-65` FHIR provenance [32], multi-method autorater skepticism [18], and open NHS DCB0129 hazards ensures clinicians, pharmacists, and Link Workers retain full authority.

Data, Code, and Cryptographic Receipt Availability

All source code, Guppy 1.0 HUGR binaries, and nine SHA-256-sealed JSON receipts across Quantinuum Nexus, Haiqu v1.7.1, OpenJij, HQS, `QProtSim`, Modal Cloud, Hugging Face, `ML-DSA-65`, and TypeSafe Jev 1.14 are openly available at https://arunnadarasa.github.io/quantum-social-prescribing/ and https://huggingface.co/socialprescribing/quantinuum-social-prescribing.

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